EXPERIMENT 21

AIM

Extract recently accessed files from a simulated OS activity log, sort them in reverse chronological order and optionally filter by user.

ALGORITHM

1. Store simulated OS activity records.
2. Convert timestamps into datetime values.
3. Sort records from newest to oldest.
4. Filter records for the selected user.
5. Display timestamp and filename.
6. Use the ordered result to identify recent activity.
7. Stop.

CODE

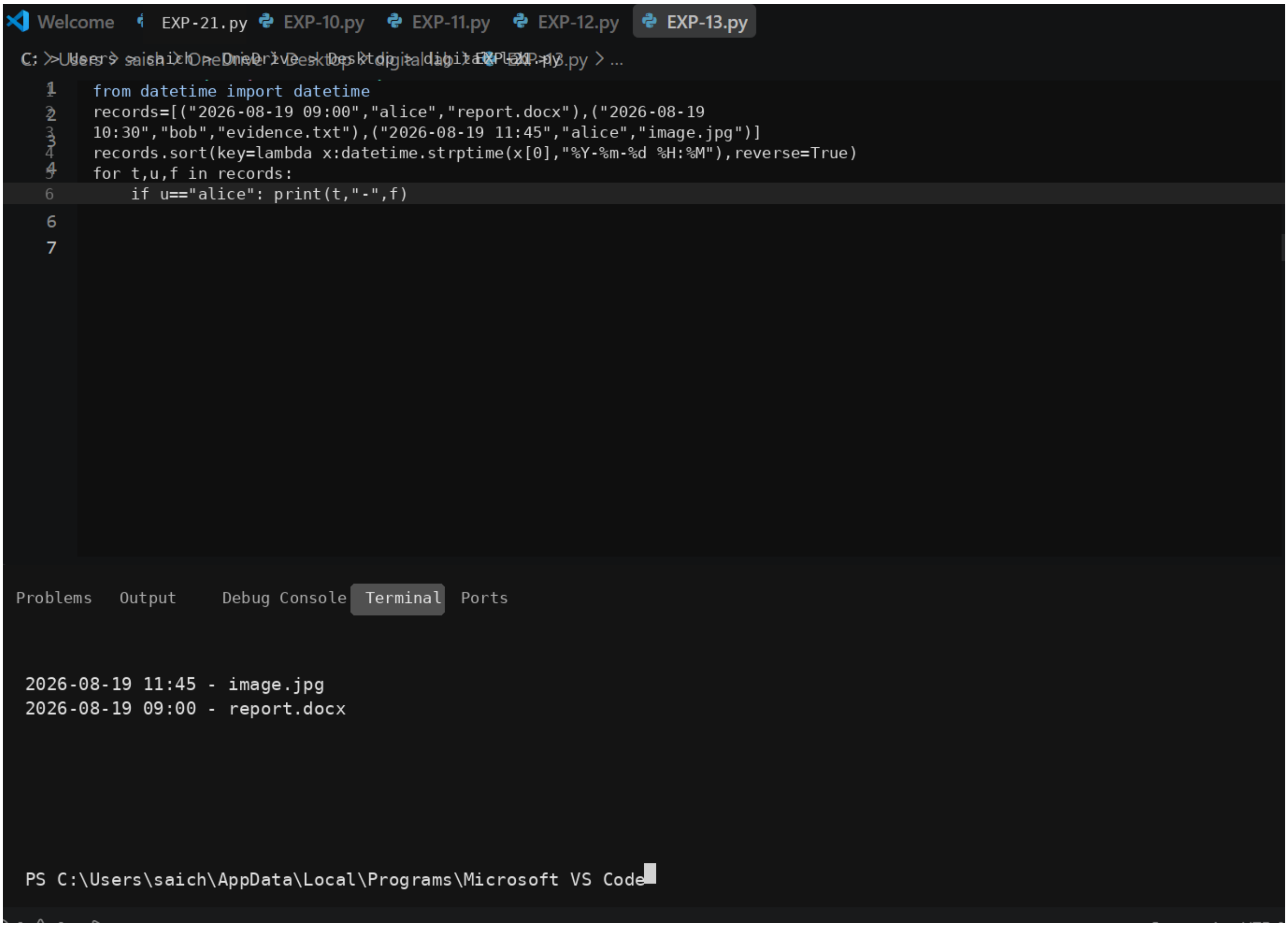
```
from datetime import datetime

records = [
    ("2026-08-19 09:00", "alice", "report.docx"),
    ("2026-08-19 10:30", "bob", "evidence.txt"),
    ("2026-08-19 11:45", "alice", "image.jpg")
]

records.sort(
    key=lambda item: datetime.strptime(item[0], "%Y-%m-%d %H:%M"),
    reverse=True
)

for timestamp, user, filename in records:
    if user == "alice":
        print(timestamp, "-", filename)
```

OUTPUT



RESULT

The program identifies recently accessed files and arranges them in reverse chronological order with a username filter.